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Longitudinal Multimodal Cardiovascular Patient-State Representation

A Public Scientific Framework for Temporal Coherence, Evidence Integration, Provenance, Uncertainty, and Clinician-Interpretable Cardiovascular Trajectories

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Appropriate for this public paper	Excluded from this public paper
Clinical and scientific rationale; high-level definitions; public standards context; research hypotheses; general validation principles; limitations.	Source code; model weights; prompts; internal algorithms; proprietary scoring formulas; production thresholds; detailed state-transition rules.
General multimodal data domains and temporal challenges.	Internal data schemas; proprietary mappings; interface specifications; system topology; security design; customer or tenant configurations.
Conceptual examples and public research questions.	Nonpublic datasets; confidential partner information; unreleased patent claims; trade-secret implementation details; deployment procedures.

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0.1-P	Public Scientific Edition	2026	Initial public-safe framework defining the scientific problem, public principles, validation domains, and research agenda.

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Table of Contents

1. Abstract
2. Introduction
3. Why Cardiovascular Snapshots Are Insufficient
4. Definition of Cardiovascular Patient-State Representation
5. Public Framework Principles
6. Multimodal Cardiovascular Evidence Domains
7. Temporal Architecture and Clinical Event Context
8. Semantic Interoperability and Cross-Modal Alignment
9. Provenance, Lineage, and Data Integrity
10. Missingness, Conflict, and Uncertainty
11. Public Conceptual Representation Layers
12. State Updating, Transition, and Reversibility
13. Clinician Interpretation and Human Oversight
14. Illustrative Cardiovascular Applications
15. Validation and Evidence Framework
16. Lifecycle Governance and Monitoring
17. Standards and Regulatory Context
18. Research Agenda
19. Limitations
20. Conclusions
21. References
Appendix A - Public Patient-State Reporting Checklist
Appendix B - Public-Safe Validation Matrix

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1. Abstract

Cardiovascular care produces heterogeneous evidence across electrocardiographic waveforms, cardiovascular imaging, hemodynamics, laboratory measurements, medications, procedures, implanted devices, wearable and remote monitoring, clinical observations, genomic and molecular information, and patient-reported outcomes. These data are typically acquired at different times, for different clinical purposes, through different systems, and with different levels of quality and provenance. A collection of measurements is therefore not automatically a coherent representation of the patient's cardiovascular condition.

This white paper proposes Longitudinal Multimodal Cardiovascular Patient-State Representation as a public scientific framework for organizing cardiovascular evidence around the evolving patient rather than around isolated tests or encounters. A patient-state representation is defined here as a time-indexed, evidence-bound, provenance-preserving, uncertainty-aware, and clinician-interpretable account of what is known, what has changed, what interventions have occurred, what remains unresolved, and which observations remain applicable to the current clinical question.

The framework distinguishes observation time, physiologically relevant time, documentation time, result-availability time, and intervention time; separates direct observations from derived interpretations; preserves modality-specific context; identifies conflicts and material missingness; and requires traceability from any state summary back to its source evidence. HL7 FHIR and DICOM provide important standards for representing and exchanging clinical and imaging information, but no single standard alone creates a clinically meaningful longitudinal patient state.[1-4]

The paper outlines public principles, data domains, temporal and provenance requirements, validation domains, lifecycle governance questions, and a research agenda. It intentionally does not disclose xBxBio implementation details. The framework remains proposed and testable. Empirical work is required to determine whether patient-state representation improves clinical understanding, validation reliability, workflow performance, or patient outcomes.

2. Introduction

Clinical care is longitudinal, but most digital records are organized as transactions: a visit, an image, a waveform, a laboratory result, a prescription, a procedure, or a note. The clinician must mentally reconstruct the patient's evolving condition across these fragments. This reconstruction becomes more difficult as data volume, modality diversity, care-team size, and care-setting transitions increase.

Cardiovascular medicine makes the problem especially visible. Electrical, mechanical, structural, hemodynamic, metabolic, behavioral, and molecular signals may each describe a different aspect of cardiovascular function. Their meaning depends not only on measured values but also on acquisition context, temporal relationships, treatment exposure, physiologic state, and intervening events. An echocardiogram obtained before a major medication change, an ECG acquired during an acute illness, and a wearable trend collected after discharge cannot be interpreted as though they were simultaneous observations of one stable state.

Research in longitudinal EHR representation, temporal phenotyping, multimodal learning, and cardiovascular foundation models shows that integrating multiple information sources can improve analytical capability, but these methods also reveal challenges involving long sequences, irregular intervals, modality alignment, missingness, and generalizability.[7-12] The scientific problem is

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therefore not merely to combine data. It is to preserve enough context to determine what the combined information means at a particular point in the patient journey.

This paper positions patient-state representation as an intermediate scientific object between raw clinical data and clinically consequential interpretation. It is not a diagnosis, a risk score, a digital twin, or a treatment recommendation. It is a structured representation of evidence and uncertainty that may support those downstream activities when appropriately validated.

2.1 Central Research Question

How can heterogeneous cardiovascular evidence be organized over time so that clinically meaningful changes, interventions, uncertainty, provenance, and unresolved conflicts remain visible and independently reviewable?

2.2 Scope

- Longitudinal representation of cardiovascular evidence across multiple clinical and patient-generated modalities.
- Temporal coherence, event context, provenance, missingness, uncertainty, and human reviewability.
- Public scientific requirements that can be studied independently of any specific proprietary implementation.
- Use in research, clinical decision-support evaluation, monitoring, and future patient-specific modeling.

2.3 Out of Scope

- A production data model, proprietary xBxBio architecture, or complete interface specification.
- A universal cardiovascular ontology or a replacement for FHIR, DICOM, LOINC, SNOMED CT, UCUM, or other standards.
- An autonomous diagnostic or treatment system.
- A claim that a represented patient state is complete, correct, or clinically useful without validation.
- A public disclosure of patent-sensitive algorithms, thresholds, scoring logic, or deployment methods.

3. Why Cardiovascular Snapshots Are Insufficient

A snapshot describes selected evidence at a particular moment or encounter. It may be clinically useful, but it can obscure trajectory. Two patients with the same current ejection fraction, blood pressure, troponin level, or rhythm classification may have very different clinical meanings if one is stable and the other is deteriorating after an intervention.

Snapshot interpretation can also conceal the relationship between observation and action. A laboratory value can precede or follow a medication change; an imaging finding can be the trigger for a procedure or the result of it; a wearable signal can reflect baseline physiology or acute illness. Without event context, a system may detect statistical difference while missing clinical significance.

The American Heart Association has emphasized both the promise of AI across cardiovascular data domains and the continuing need to demonstrate meaningful clinical value, implementation quality, bias mitigation, and improved outcomes.[5,6] Longitudinal patient-state representation addresses one part of that challenge by requiring clinically relevant chronology and evidence context before information is summarized or modeled.

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3.1 Common Snapshot Failure Modes

Failure mode	Potential consequence
Encounter isolation	Important changes between visits are hidden or disconnected from their causes.
Temporal flattening	Pre-intervention and post-intervention data are treated as comparable without qualification.
Modality silos	ECG, imaging, laboratory, medication, and symptom evidence are interpreted independently.
Last-value dominance	The latest value is assumed to be the best representation regardless of quality or context.
Documentation-time confusion	The time a result was entered is mistaken for the time the physiology was observed.
Unresolved conflict suppression	Contradictory sources are collapsed into a single summary without preserving disagreement.
Missingness invisibility	Absent evidence is treated as normal or irrelevant even when it could change interpretation.

4. Definition of Cardiovascular Patient-State Representation

Proposed definition: A longitudinal multimodal cardiovascular patient-state representation is a time-indexed, evidence-bound, provenance-preserving, uncertainty-aware, and clinician-interpretable representation of the patient's cardiovascular condition, trajectory, interventions, and unresolved evidence at a defined clinical time anchor.

The phrase patient state does not imply perfect knowledge of the patient. It represents the best defensible account supported by the available evidence, with limitations preserved rather than erased. The state is expected to be revised as new evidence becomes available or previous evidence is corrected, reinterpreted, or invalidated.

4.1 Distinctions

Concept	How it differs from patient-state representation
Raw data collection	Stores observations but may not establish temporal or clinical relationships.
Clinical summary	Condenses information, often without retaining explicit source lineage or uncertainty.
Phenotype	Characterizes a pattern or subgroup; may be static or task-specific.
Risk score	Estimates an outcome probability for a defined endpoint and horizon.
Digital twin	May simulate patient-specific physiology or interventions; patient-state representation can be an

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Concept	How it differs from patient-state representation
	input but is not equivalent.
Diagnosis	A clinical conclusion; the patient state may contain evidence relevant to diagnosis but does not replace clinician judgment.

5. Public Framework Principles

Principle	Public scientific requirement
Patient-centered	The representation is organized around the patient and clinical question rather than around the source system.
Time-indexed	Every material observation, interpretation, intervention, and uncertainty is related to a clinically meaningful time anchor.
Multimodal	Electrical, structural, functional, hemodynamic, biochemical, behavioral, molecular, and narrative evidence may be represented without assuming equivalence.
Provenance-preserving	Source, method, timing, transformation, responsible actor, and version information remain traceable.
Intervention-aware	Medication, procedure, device, care-setting, and behavioral changes are represented as potential causes of state transition.
Uncertainty-aware	Missingness, conflict, data quality, model uncertainty, and clinical ambiguity remain explicit.
Clinician-interpretable	The representation supports independent review of material evidence and does not require blind reliance on a composite score.
Revisable	New evidence may update the state; corrections and reinterpretations remain auditable.
Purpose-bounded	A state representation is evaluated for a defined clinical or research purpose rather than presumed universally valid.
Outcome-linked	Where appropriate, state changes and interventions can be studied in relation to subsequent outcomes without implying causation.

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6. Multimodal Cardiovascular Evidence Domains

A public framework should identify the evidence domains that may contribute to cardiovascular state while preserving modality-specific meaning. Inclusion of a domain does not imply that every patient or use case requires every modality.

Evidence domain	Examples and context
Patient and care context	Age, sex-related variables where clinically relevant, comorbidities, care setting, encounter type, clinical question, goals of care, and relevant social context.
Symptoms and examination	Dyspnea, chest discomfort, palpitations, syncope, edema, functional status, physical findings, and clinician observations.
Electrocardiography and waveforms	Resting ECG, ambulatory ECG, telemetry, intracardiac signals, heart rate, rhythm, intervals, morphology, and signal-quality context.
Cardiovascular imaging	Echocardiography, stress imaging, cardiac CT, CMR, nuclear imaging, vascular imaging, measurements, reports, and acquisition context.
Hemodynamics and physiology	Blood pressure, filling pressures, flow, gradients, oxygenation, exercise response, autonomic markers, and invasive or noninvasive measurements.
Laboratory and biomarkers	Renal function, electrolytes, lipids, glucose measures, natriuretic peptides, troponin, inflammatory markers, and other relevant biomarkers.
Medications and therapies	Orders, dispensing, administration, adherence evidence, dose changes, therapeutic response, contraindications, and adverse effects.
Procedures and devices	Ablation, revascularization, valve intervention, surgery, implanted devices, device interrogation, and post-procedure status.
Wearables and remote monitoring	Consumer or medical-grade signals, activity, sleep, pulse, rhythm, weight, blood pressure, transmission quality, and device provenance.
Genomic and molecular evidence	Genetic variants, family history, molecular markers, and interpretation status where relevant and appropriately governed.
Patient-reported outcomes	Symptoms, quality of life, treatment burden, preferences, function, and self-reported changes.
Clinical narrative	Assessment, differential diagnosis, rationale, uncertainty, care plan, and contextual information not fully captured in structured fields.

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7. Temporal Architecture and Clinical Event Context

FHIR distinguishes the physiologically relevant time of an Observation from the time a result version is issued, and it can link observations to encounter context.[2] This distinction is fundamental to patient-state representation. A result may be documented after the condition it describes, and a note may refer retrospectively to an earlier episode.

A public temporal model should therefore preserve more than one timestamp when clinically material. It should also recognize intervals, repeated measurements, episodes, baselines, and intervention boundaries. Clock time alone is insufficient when a major procedure, medication change, acute illness, or care transition changes the meaning of prior evidence.

Temporal representation should not imply that all modalities update at the same rate. ECG rhythm may change within seconds; hemodynamics within minutes; biomarkers over hours; imaging over days or months; genomic evidence may be stable while its interpretation changes. The representation should preserve these different temporal characteristics.

7.1 Public Temporal Elements

Temporal element	Meaning
Clinical time anchor	The point or interval for which the patient state is being represented.
Observation/effective time	When the measured or asserted clinical condition was physiologically relevant.
Acquisition time	When the signal, image, specimen, or patient report was obtained.
Result-availability time	When the information became available to the care team or system.
Documentation time	When the information was entered, signed, amended, or imported.
Intervention time	When medication, procedure, device, or care-setting changes occurred.
Validity interval	The period during which evidence may remain applicable for the defined purpose.
Event relationship	Before, during, after, triggered by, response to, or unrelated to a clinically material event.

7.2 Event Boundaries

- Acute presentation or hospitalization.
- New arrhythmia or rhythm conversion.
- Medication initiation, discontinuation, or dose change.
- Procedure, surgery, or device implantation.
- Change in volume status, renal function, or systemic illness.
- Change in care setting or monitoring intensity.
- New diagnostic evidence that materially revises the clinical interpretation.

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8. Semantic Interoperability and Cross-Modal Alignment

FHIR is a standard for health data exchange and provides resources for observations, encounters, reports, conditions, procedures, medications, devices, and provenance.[1-3] DICOM is the standard for communication and management of medical imaging information and related data, including cardiology and waveforms.[4] These standards provide essential building blocks, but conformance to a standard does not by itself guarantee that two systems interpret data identically or that a coherent patient state has been constructed.

Cross-modal alignment requires preservation of clinical meaning across coding systems, units, measurement methods, body sites, acquisition protocols, and reporting conventions. A left-ventricular ejection fraction derived by different modalities or methods may not be directly interchangeable. A rhythm diagnosis extracted from a note is not equivalent to a verified waveform interpretation. A medication order is not the same as confirmed administration or adherence.

The public scientific requirement is semantic transparency: the representation should make visible what an item means, how it was produced, and whether comparisons across time or modality are valid for the intended purpose.

8.1 Alignment Questions

- Are terms coded consistently and traceably to recognized vocabularies where feasible?
- Are units explicit and normalized without losing original values?
- Are modality-specific methods and measurement conventions preserved?
- Are duplicate, amended, preliminary, and final results distinguished?
- Are derived values distinguishable from directly measured values?
- Are data linked to the correct patient, encounter, body structure, device, and clinical question?
- Can a reviewer determine whether two values are genuinely comparable?

9. Provenance, Lineage, and Data Integrity

FHIR Provenance supports representation of the entities, agents, activities, versions, and transformations associated with clinical information.[3] In a longitudinal multimodal state, provenance is not optional decoration. It is part of the scientific meaning of the evidence.

A patient-state summary may include direct measurements, clinician interpretations, imported records, normalized units, extracted concepts, algorithm-derived features, and generated summaries. Each layer can introduce error or ambiguity. The representation should therefore support traceability from a summarized assertion back to the original source and the transformations applied to it.

Data integrity also includes identity integrity, chronology integrity, version integrity, and correction history. A result linked to the wrong patient, the wrong encounter, or the wrong time can be more dangerous than a missing result because it appears complete.

9.1 Minimum Public Provenance Expectations

- Source organization and source system where available.
- Patient and encounter linkage.
- Acquisition method, device, or modality where clinically relevant.
- Original observation time and result/documentation time.
- Original value, unit, and later normalization or transformation.

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- Responsible clinician, organization, device, or software agent where appropriate.
- Preliminary, final, corrected, amended, or superseded status.
- Derivation lineage for summaries, features, or AI-generated content.
- Version of the model or rule set used for any derived interpretation.

10. Missingness, Conflict, and Uncertainty

Missing evidence is not uniformly negative, neutral, or random. A test may be absent because it was not clinically indicated, inaccessible, deferred, refused, performed elsewhere, or not captured by the receiving system. These explanations carry different meanings and potential biases.

Conflicting modalities should not automatically be averaged or resolved by model confidence. Disagreement can indicate measurement error, temporal change, method difference, physiologic complexity, or an unresolved clinical question. A defensible patient state should preserve material conflict and the evidence needed for adjudication.

Uncertainty includes measurement error, missingness, contradictory evidence, model uncertainty, interpretive ambiguity, and uncertainty about whether prior evidence remains current. These dimensions should be represented in ways that clinicians and researchers can understand and validate.

10.1 Public Uncertainty Categories

Category	Illustrative meaning
Measurement uncertainty	Noise, limited image quality, sampling variability, device limitations, or assay uncertainty.
Temporal uncertainty	Unclear timing, unknown onset, or uncertainty about whether evidence remains current.
Provenance uncertainty	Source, identity, transformation, or version cannot be established adequately.
Semantic uncertainty	Meaning differs across codes, methods, reports, or natural-language interpretations.
Clinical uncertainty	Evidence supports multiple plausible interpretations.
Model uncertainty	The computational model has limited confidence or is outside its evaluated context.
Missingness uncertainty	The reason for absent evidence is unknown or potentially informative.

11. Public Conceptual Representation Layers

The following layers are presented as public scientific requirements, not as an xBxBio implementation specification.

Public conceptual layer	Purpose
1. Identity and context	Patient, encounter, care setting, clinical question,

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Public conceptual layer	Purpose
	intended use, and decision horizon.
2. Source evidence	Time-stamped multimodal observations with original method, units, quality, and provenance.
3. Event and intervention history	Medications, procedures, devices, acute episodes, and other events capable of changing state.
4. Temporal relationships	Before/after relationships, intervals, trends, episodes, baselines, and validity conditions.
5. Derived clinical features	Transparent, traceable summaries or measurements derived from source evidence.
6. Uncertainty and conflict	Material missingness, discordance, data-quality limitations, and unresolved questions.
7. Clinician assertions	Documented interpretations, differential diagnosis, rationale, overrides, and adjudications.
8. Outcome and response	Subsequent events, treatment response, adverse outcomes, and follow-up evidence.

11.1 Observation State, Inference State, and Action State

A useful research distinction is between what has been observed, what has been inferred, and what action is being considered. Collapsing these into one composite state can obscure uncertainty and causality. The observation state should preserve source evidence; the inference state should identify derived interpretation and its basis; the action state should identify the clinical workflow consequence without implying that an inference automatically determines care.

12. State Updating, Transition, and Reversibility

A patient-state representation must change when clinically material evidence changes. The update process should preserve the previous state, the trigger for transition, and the evidence responsible for the update. This is especially important when later evidence overturns an earlier interpretation.

State transition is not synonymous with disease progression. A represented state can change because of improved data quality, correction of patient identity, new context, reinterpretation, or resolution of uncertainty. The representation should distinguish physiologic change from knowledge change.

Reversibility is an auditability requirement. A reviewer should be able to reconstruct what the system and clinician knew at a prior time, not merely view the latest overwritten state.

12.1 Public Transition Triggers

- New observation or diagnostic report.
- New or changed medication, procedure, or device information.
- Clinically material event or care transition.
- Correction, amendment, or invalidation of prior evidence.
- Change in data quality or provenance status.
- Clinician adjudication or reinterpretation.

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- New outcome or treatment-response information.
- Change in intended use or clinical question.

13. Clinician Interpretation and Human Oversight

A patient-state representation should support, not replace, clinician judgment. FDA's 2026 Clinical Decision Support Software guidance emphasizes that, for applicable non-device CDS, healthcare professionals must be able to independently review the basis of recommendations rather than rely primarily on the software.[18] Patient-state representation can support this goal when it preserves required inputs, source evidence, knowns, unknowns, timing, and derivation.

Human oversight is meaningful only when the representation is understandable in the available workflow. A technically complete lineage graph that cannot be interpreted under clinical time pressure may not provide effective oversight. The representation should therefore support layered review: a concise state summary, visible material limitations, and access to underlying evidence when needed.

Clinicians should also be able to challenge, correct, or qualify the represented state. Overrides and adjudications should be recorded rather than silently replacing the prior system state.

13.1 Clinician Review Questions

- What clinical time does this state represent?
- Which sources materially support the state?
- What changed since the prior state, and why?
- What interventions occurred between relevant observations?
- Which evidence is missing, stale, low quality, or conflicting?
- Which content is directly observed versus derived or generated?
- Can the underlying source evidence be independently reviewed?
- What assumptions limit use of the represented state?

14. Illustrative Cardiovascular Applications

The following examples are conceptual and are not validated clinical decision rules.

Illustrative use case	Patient-state purpose
Heart failure trajectory	Represent symptoms, weight, blood pressure, renal function, natriuretic peptides, imaging, medication changes, device data, and hospital events around a defined time anchor rather than as an unstructured list.
Atrial fibrillation journey	Relate rhythm evidence, burden, symptoms, anticoagulation, rate/rhythm-control therapy, procedures, monitoring quality, and outcome events over time.
Ischemic heart disease	Align symptoms, ECG, biomarkers, imaging, angiography, intervention, medication, and follow-up while preserving pre- and post-revascularization context.
Cardiomyopathy	Integrate ECG, echo, CMR, genetics, family history,

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Illustrative use case	Patient-state purpose
	arrhythmia, hemodynamics, and longitudinal structural change without collapsing discordant evidence.
Valvular disease	Represent severity measurements, symptoms, ventricular response, procedural timing, device status, and post-intervention recovery across modality and time.
Cardio-oncology	Relate baseline cardiovascular risk, cancer therapy exposure, biomarkers, imaging, rhythm, symptoms, and long-term treatment effects; AHA guidance highlights the importance of longitudinal individualized risk in this domain.[20]
Remote monitoring	Connect wearable or implanted-device signals to the current care context, device provenance, symptom reports, and confirmatory clinical evidence.

15. Validation and Evidence Framework

Validation must assess whether the representation faithfully preserves clinically relevant evidence and relationships, not merely whether a downstream prediction performs well. A patient-state representation can be internally consistent yet clinically misleading if important events, conflicts, or provenance limitations are omitted.

TRIPOD+AI provides reporting guidance for prediction-model studies, and DECIDE-AI addresses early clinical evaluation of AI decision support in real workflows.[13,14] SPIRIT-AI and CONSORT-AI extend clinical-trial protocol and reporting guidance for AI interventions.[15,16] Patient-state representation studies should use these frameworks where relevant while adding representation-specific validation targets.

Validation should be use-case-specific. The evidence required for retrospective phenotyping may differ from that required for time-sensitive clinical support, and the tolerable uncertainty may differ by consequence.

15.1 Proposed Validation Domains

Validation domain	Example evaluation question
Source fidelity	Are source values, methods, units, quality indicators, and identities preserved accurately?
Temporal fidelity	Are observation, intervention, and documentation times represented correctly?
Event coherence	Are pre-event and post-event data separated and related appropriately?
Cross-modal fidelity	Are modality-specific meanings preserved and comparable only when justified?

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Validation domain	Example evaluation question
Provenance completeness	Can material assertions be traced to source evidence and transformations?
Missingness behavior	Does the representation identify clinically material absent evidence without treating all missingness identically?
Conflict preservation	Are disagreements retained until appropriately adjudicated?
Update correctness	Does new evidence produce the intended state change while preserving prior-state history?
Clinician interpretability	Can clinicians understand state basis, limitations, changes, and uncertainties?
External transportability	Does representation quality persist across sites, populations, devices, and workflows?
Equity and subgroup effects	Do data access, missingness, and modality availability create unequal representation quality?
Clinical utility	Does use of the representation improve decisions, workflow, safety, or outcomes in prospective evaluation?

15.2 Representation Ground Truth

There may be no single perfect ground truth for a complex cardiovascular state. Validation may therefore require adjudication by multidisciplinary experts using predefined evidence rules, explicit disagreement handling, and inter-rater reliability assessment. The adjudication process itself should be transparent and reproducible.

15.3 Counterfactual and Perturbation Testing

- Remove a clinically material observation and test whether uncertainty or missingness becomes visible.
- Shift an observation across an intervention boundary and test whether the representation changes appropriately.
- Introduce conflicting modality evidence and verify that disagreement is preserved.
- Change provenance or identity linkage and verify that the state is qualified or invalidated.
- Correct a prior result and verify that historical and current states remain reconstructable.
- Test population, site, device, and workflow shifts for degradation in representation fidelity.

16. Lifecycle Governance and Monitoring

Patient-state representation is a lifecycle function. Data sources, interfaces, clinical workflows, devices, coding practices, and computational models change over time. A representation that was validated under one combination of sources and workflows may not remain valid after material change.

FDA's 2025 draft guidance for AI-enabled device software functions describes total-product-lifecycle risk-management considerations, and the 2025 final PCCP guidance addresses planned modifications for applicable AI-enabled devices.[17,19] NIST's AI RMF provides a voluntary Govern-Map-Measure-

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Manage structure for AI risk.[21] These sources support the principle that representation quality, change, and risk should be monitored throughout the lifecycle.

Monitoring should include upstream data availability, timestamp quality, provenance failures, modality drift, missingness patterns, state-update frequency, clinician corrections, and incidents in which the represented state contributed to misunderstanding or delay.

16.1 Public Monitoring Indicators

- Rate of missing or temporally ambiguous evidence by source and population.
- Frequency of patient, encounter, and device identity-linkage problems.
- Rate of conflicting multimodal evidence and time to adjudication.
- Frequency of clinician correction, override, or qualification.
- Changes in modality availability, acquisition method, or coding practice.
- Population and site differences in representation completeness.
- Downstream model performance stratified by representation quality.
- Safety events, near misses, or workflow delays associated with state interpretation.

17. Standards and Regulatory Context

The framework is not presented as a new interoperability standard or regulatory requirement. It is a public scientific model that can be mapped to existing standards and guidance where appropriate.

Source	Relevance
HL7 FHIR R5	Provides exchange resources and temporal/context fields for observations, reports, encounters, procedures, medications, devices, and provenance. It is a building block, not a complete patient-state methodology.[1-3]
DICOM current edition	Supports communication and management of medical imaging information and related data, including cardiology and waveforms. DICOM conformance does not by itself guarantee semantic or clinical interoperability.[4]
AHA scientific statements	Identify opportunities and persistent evidence, implementation, bias, workflow, and outcome challenges for AI in cardiovascular medicine and imaging.[5,6]
FDA AI-enabled device guidance	Provides lifecycle and submission considerations for applicable AI-enabled device software functions; the January 2025 lifecycle guidance remains draft as of this document date.[17]
FDA Clinical Decision Support guidance	Clarifies FDA thinking regarding device and certain non-device CDS functions and independent review of recommendation basis.[18]
FDA PCCP guidance	Provides final recommendations for planned modifications to applicable AI-enabled device software functions.[19]
NIST AI RMF	Voluntary framework for managing AI risk and trustworthiness; NIST states AI RMF 1.0 is under revision in 2026.[21]
WHO multimodal-model guidance	Addresses ethics and governance of large multimodal

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Source	Relevance
	models in health, including stakeholder engagement, reliability, data quality, bias, and post-release oversight. [22]

18. Research Agenda

Priority scientific questions include:

- What minimum evidence elements are required to represent cardiovascular state for specific clinical purposes?
- How should observation time, intervention time, documentation time, and result-availability time be reconciled?
- How should a representation distinguish physiologic change from change in knowledge or data quality?
- Which missingness patterns are clinically material, and how should their meaning be represented?
- How can conflicting modalities be preserved without overwhelming the clinician?
- What metrics best measure source, temporal, cross-modal, and provenance fidelity?
- How should clinicians adjudicate a reference patient state, and how reproducible is that adjudication?
- How does representation quality vary across populations, sites, devices, and levels of access to advanced testing?
- Does multimodal representation add value when some modalities are unavailable at inference time?
- Can patient-state representation improve calibration, transportability, or interpretability of downstream AI?
- Can a representation support safe simulation or digital-twin research without overstating physiological fidelity?
- Does prospective use improve clinical decisions, care coordination, safety, workflow, or patient outcomes?

18.1 Suggested Research Program

Phase	Public research objective
1. Specification	Define use-case-specific state elements, time anchors, provenance, missingness, conflict, and clinician-review requirements.
2. Retrospective construction	Build representations from retrospective records and measure source, temporal, and cross-modal fidelity.
3. Expert adjudication	Compare representation content with multidisciplinary expert review and measure disagreement.
4. Silent prospective evaluation	Generate patient states without influencing care to study update behavior and data limitations.
5. Human-factors evaluation	Assess clinician understanding, correction, workload,

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Phase	Public research objective
	trust calibration, and workflow fit.
6. Prospective impact evaluation	Where justified, test effects on decision quality, safety, resource use, and outcomes.
7. Lifecycle monitoring	Evaluate drift, change control, subgroup effects, and incident-linked failures after deployment.

19. Limitations

- The proposed framework is conceptual and has not been independently validated as a general patient-state representation methodology.
- Cardiovascular state is not fully observable, and expert interpretations can differ even when reviewing the same evidence.
- Multimodal data availability is unequal across patients, sites, regions, and care settings, which can create representation bias.
- High-level public principles do not resolve implementation questions involving data models, algorithms, thresholds, performance, or cybersecurity.
- Interoperability standards support exchange but do not guarantee consistent clinical meaning or data quality.
- A richer representation can increase cognitive burden if interface design and workflow integration are poor.
- Associations between represented state, interventions, and outcomes do not establish causation.
- Regulatory interpretation depends on actual intended use, functions, claims, users, and jurisdiction.
- References and regulatory status must be reverified immediately before any later final publication.

20. Conclusions

Cardiovascular care cannot be reduced safely to a sequence of disconnected snapshots. Clinically meaningful interpretation depends on trajectory, intervention history, modality context, provenance, uncertainty, and the distinction between when information was observed and when it became available.

Longitudinal Multimodal Cardiovascular Patient-State Representation is proposed as a public scientific framework for preserving those relationships. The framework treats patient state as a revisable, evidence-bound representation rather than a single score or autonomous clinical conclusion.

The scientific value of the framework will depend on rigorous validation: source fidelity, temporal fidelity, cross-modal integrity, missingness and conflict behavior, clinician interpretability, external transportability, fairness, workflow performance, and prospective clinical utility.

This paper establishes the public research question and evidence boundary. It does not disclose how xBxBio may implement these principles. The next step is independent critique and empirical study, not stronger claims.

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Appendix A - Public Patient-State Reporting Checklist

Checklist domain	Public reporting expectation
Purpose and time anchor	The clinical/research purpose and represented time or interval are stated.
Patient and encounter identity	Evidence is linked to the correct patient and relevant encounter or care context.
Source evidence	Material evidence domains and original sources are identifiable.
Temporal relationships	Observation, acquisition, result, documentation, and intervention times are distinguished where material.
Interventions	Medication, procedure, device, and care-setting changes relevant to interpretation are visible.
Provenance	Source, method, version, transformation, and responsible actor are traceable where applicable.
Direct vs derived content	Measured, clinician-interpreted, and computationally derived information are distinguishable.
Missingness	Clinically material absent evidence and known reasons are represented.
Conflict	Material disagreement between sources or modalities remains visible until adjudicated.
Uncertainty	Measurement, temporal, semantic, clinical, provenance, and model uncertainty are represented as relevant.
State change	Material changes from the prior state and their triggers are identifiable.
Clinician review	A qualified clinician can review the basis, limitations, and underlying evidence.
Version history	Prior states, corrections, amendments, and overrides remain auditable.
Validation scope	The intended use, population, setting, data sources, and validation limits are stated.

Appendix B - Public-Safe Validation Matrix

Domain	Illustrative test	Potential failure signal
Temporal fidelity	Compare represented event order and intervals with adjudicated source records.	Pre/post intervention data merged incorrectly; stale evidence treated as current.
Source fidelity	Reconcile displayed values and methods to original records.	Unit, value, method, or identity mismatch.
Cross-modal alignment	Review comparable and non-	Inappropriate equivalence or loss of

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Domain	Illustrative test	Potential failure signal
	comparable measurements across modalities.	modality context.
Provenance	Trace summarized assertions to source and transformations.	Untraceable or version-ambiguous content.
Missingness	Remove material evidence and evaluate state qualification.	Missing data hidden or incorrectly interpreted as normal.
Conflict	Introduce discordant evidence and review representation behavior.	Conflict collapsed into false certainty.
Update behavior	Add, correct, or invalidate evidence and review state history.	Prior state overwritten; incorrect transition.
Clinician interpretation	Test understanding under realistic workflow conditions.	Basis, limitations, or changes misunderstood.
External transportability	Evaluate across institutions, devices, and populations.	Representation quality degrades outside development context.
Equity	Stratify completeness and errors by population and access patterns.	Unequal representation quality or missingness burden.

About xBxBio

xBxBio advances Connected Cardiovascular Intelligence through the responsible integration of longitudinal cardiovascular information, multimodal clinical data, interoperability, clinical artificial intelligence, data integrity, provenance, governance, validation, and human clinical oversight.

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